Capacity-feasible clustering on neutral atoms

Pasqal Track A

Assign each customer once. Respect vehicle capacity.
Minimize pairwise distance within each group.

Main finding

The atomic layout can change a correct model's lowest-energy answer into an infeasible assignment.

Noiseless emulations throughout. Exact routing evaluates the selected clusters.

Assignment: one-hot baseline and reduced codes

Codes below select vehicles in order. Reduced assignment makes the last vehicle implicit.

Vehicles	One-hot baseline: valid codes	Reduced: valid codes
2	10, 01	1, 0
3	100, 010, 001	10, 01, 00
4	1000, 0100, 0010, 0001	100, 010, 001, 000

One-hot: one bit per vehicle

$$A\sum_{i}\Big(\sum_{k=1}^{K} x_{ik} - 1\Big)^2$$

Reduced: save one bit per customer

$$x_{iK} = 1 - \sum_{k<K} y_{ik}, \quad x_{ik} = y_{ik}\ (k < K)$$

$$K = 2:\ P_i = 0 \qquad K = 3:\ P_i = Ma_ib_i \qquad K = 4:\ P_i = M(a_ib_i + a_ic_i + b_ic_i)$$

Linear substitution keeps a QUBO. Large enough assignment penalties enforce valid codes. Capacity is separate.

Capacity slack enforces a limit; balance favors equal loads

Use either capacity model with one-hot or reconstructed assignment indicators.

Capacity slack: extra bits, exact constraint

$$B\sum_k (L_k + s_k - Q)^2, \quad s_k \geq 0$$

Nonnegative spare capacity absorbs unused load.
Overload cannot have zero penalty.

Balance: no extra bits, soft preference

$$\mu\sum_k (L_k - Q)^2, \quad L_k = \sum_i d_i x_{ik}$$

Penalizes unequal loads, including legal ones.
A better clustering cost can outweigh it.

Example: capacity 7 per vehicle	Slack penalty	Balance penalty
Loads (5, 5): feasible	0	8 × weight
Loads (6, 4): feasible	0	10 × weight
Loads (8, 2): overloaded	Positive	26 × weight

Exact constraint claim: intended ground state, valid assignments, sufficient penalties and enough slack range.

Fewer atoms change the capacity model

Textbook penalties enforce constraints. The compact model replaces capacity with load balance.

Encoding	Small	Medium	Large
Baseline: assignment + capacity slack	14	45	80
Full assignment + balance	8	30	60
Reduced assignment + balance	4	20	45

S / M / L: 4 / 10 / 15 customers, 2 / 3 / 4 vehicles, capacity 7 / 20 / 30. Ceiling: 80 atoms.

Capacity remains a soft constraint in the reduced model.

Instance generation and verification

20 problems selected before emulation. Capacity changes the best clustering in every case.

Tier	Problems	Customers	Vehicles	Capacity	Demands
Small	10	4	2	7	1–4
Medium	5	10	3	20	1–8
Large	5	15	4	30	1–10

Generate and screen

Seeded positions in a 100 by 100 square.
Reject infeasible or capacity-inactive draws.

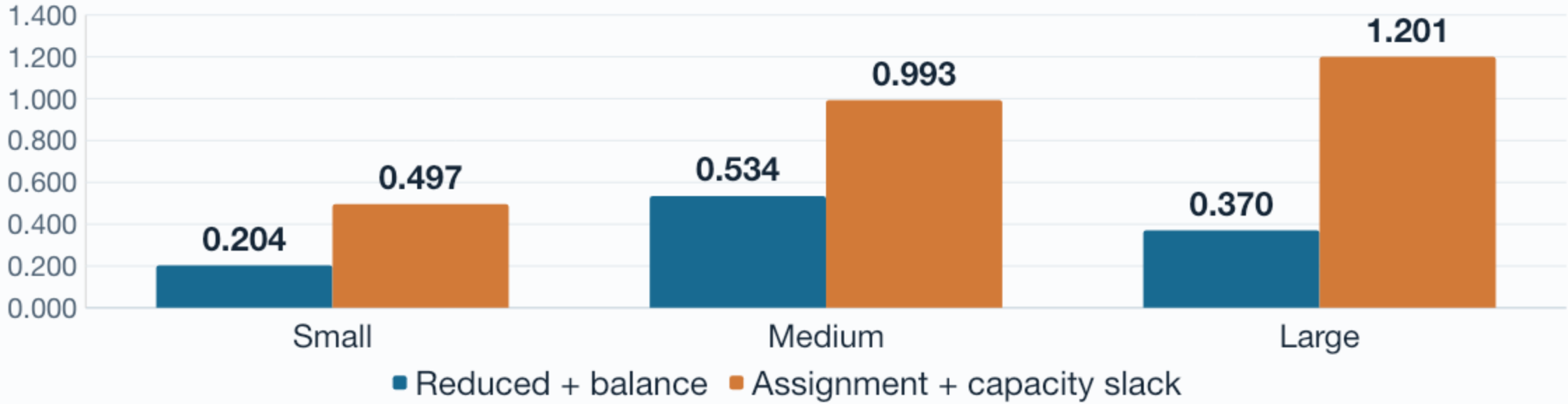
Verify each accepted instance

20 of 20 regenerate exactly.
20 of 20 independent certificates are valid.

Save seeds, certificates and demand ratios. Exact screening grows exponentially; the penalty certificate uses only demands and capacity.

Geometry restricts the interactions we can realize

All pairs interact. Distance fixes positive couplings. One global field cannot match unequal fields.



BLaDE relative pair-coefficient error after scaling. Lower is better.
10 small, 5 medium and 5 large problems; 3 layouts per problem. Includes preparation failures.

Pair distortion alone removes feasible minima

Classical solves on the same layouts, keeping the original problem and scale fixed

Problem and encoding	Distinct problems	Layouts tested	Intended model	Only pairs changed	Actual endpoint
Small slack	10	60	60 of 60	0 of 60	1 of 60
Medium reduced	5	10	10 of 10	0 of 10	0 of 10
Medium slack	5	10	10 of 10	0 of 10	0 of 5 available

Last three columns: layouts with a feasible minimum.

New BLaDE-only audit: 0 of 80 pair-only models and 0 of 72 endpoints retain feasible minima.

Table pools both placement methods. New audit: five medium problems with repeated encodings and settings; eight endpoints unavailable.

Pair distortion lowers success across three encodings

New test: three shared small problems, three encodings. No known answers used to set penalties.
Within each pair: same pulse, individual fields and scale. Only pair interactions change.

Encoding (atoms)	Feasible: exact	Feasible: geometry	Optimal: exact	Optimal: geometry
Capacity slack (14)	7.81%	1.10%	3.90%	0.55%
Full one-hot + balance (8)	21.61%	10.55%	16.00%	9.67%
Reduced + balance (4)	36.05%	24.00%	29.03%	16.38%

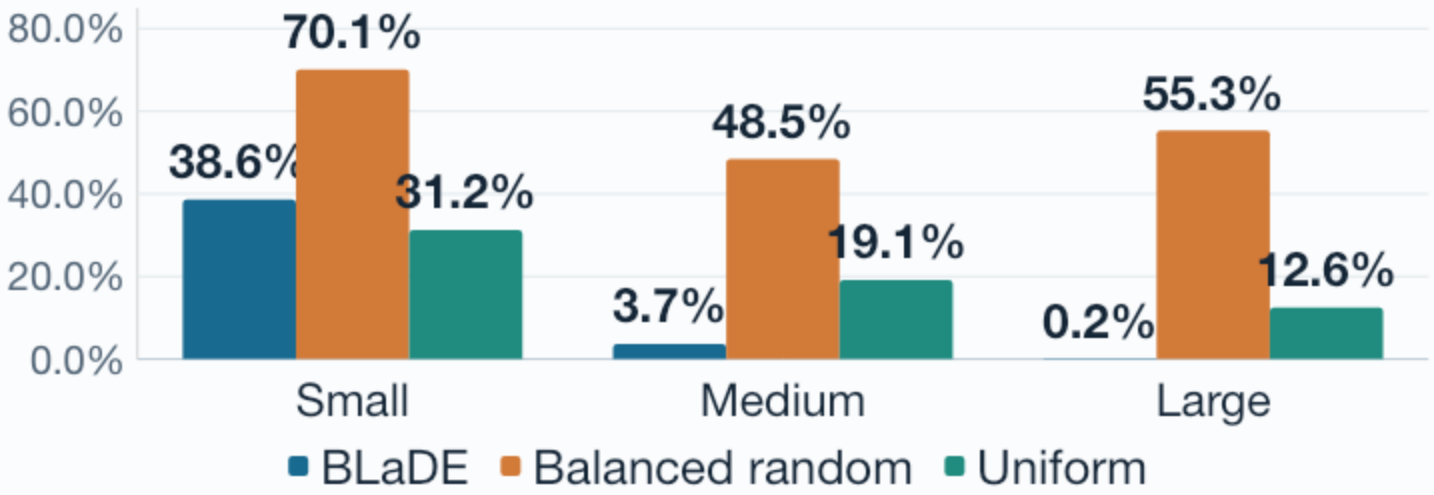
Both success rates decrease in all nine pairs. Balance also fails before embedding on one case.

Means of local state probabilities, not shot rates. One BLaDE layout per encoding and problem; ideal individual fields. All 18 timestep checks pass. Historical medium reduced cloud control: 37.025% feasible exact vs 4.020% geometric, 20,000 shots each. Full one-hot medium remains untested.

BLaDE trails simple classical sampling

Same 10 small, 5 medium and 5 large problems. Each method: 6,000 samples per problem.

Feasible samples



Original-optimal samples

Method	Small	Medium / large
BLaDE	28.68%	0% / 0%
Balanced	29.70%	0.05% / 0%
Uniform	12.72%	0.0067% / 0%

BLaDE: reduced encoding, fixed global pulse. Equal sample counts; total computation differs.

Pulse tuning exposes the embedding limit

Tuned duration, drive strength and shape, and the detuning sweep.

944 trials on 14 atoms. Per winner: 3 training, 2 validation and 5 evaluation problems.

Training interactions	Evaluation interactions	Complete-answer success
Intended	Intended	11.8–23.7%
Intended (same pulses)	Geometric	Below reporting floor
Geometric	Geometric	Up to 0.168%

Success = optimal customer assignment AND correct capacity slack bits.

Useful lesson: assignment-only tuning can hide incorrect slack.
Tuning on intended interactions did not transfer to geometry.

Penalty tuning used known optimal answers. Ideal individual fields; durations may exceed hardware limits. Ranges: three search-seed means over five cases. Reporting floor: 0.1%, not a certified bound.

Embedding fidelity is a bottleneck in these layouts

Established

Pair distortion removes feasible minima in the tested small and medium layouts.

Unresolved

Lower average coefficient error can still discard useful layouts.

Next experiment

Train on invalid low-energy candidates found without known answers. Keep the classical baseline.

Proposed follow-up only. No quantum-advantage claim. AI assisted code, analysis and writing; team review remains essential.